HOCHSCHULE
HAMM-LIPPSTADT

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Outline

I. Introduction
II. Foundations
III. Requirements
IV. Analysis & Design
V. The Development Live Cycle
VI. Implementation
VII. Verification & Validation
VIII. Summary & Outlook

SysML used as standard modeling techniques!

Introduction

- In pervious lectures
  - Systems Engineering based on INCOSE Systems Engineering Handbook
    - INCOSE Systems Engineering Handbook v. 4.0, October 2015, INCOSE-TP-2003-002-03.2.2
    - https://www.sebokwiki.org/wiki/INCOSE_Systems_Engineering_Handbook
  - Why?
    - Learn standardizes systems engineering practice
      - Handbook is consistent with ISO/IEC 15288:2008 – Systems and software engineering – System life cycle processes
- INCOSE:
  - **International Council on Systems Engineering**
  - champions the art, science, discipline, and practice of systems engineering
    - evolving Systems Engineering Body of Knowledge
      - E.g. provide a description of key process activities performed by systems engineers
    - Systems Engineering Professional (SEP) Certification program

Plan for today

- Recap plus deepen learned techniques by using a design methodology
  - That is: apply SysML to CONSENS Methodology

First part: CONSENS

- CONceptual design Specification technique for the *Engineering* of complex *Systems*
- Intelligent mechatronic systems: mechanics+electronics+software+control
- Autonomus systems
- Robotic arms
- Smart trains

6 Motivation

- Many application domains such as the automobile industry, are often based on the close interaction of mechanics, electronics and software engineering
- Communication and information technology is the enabler for advanced functionality
  - We call these: **systems with inherent partial intelligence**
- Self-optimization
  - Enables mechanical systems to react autonomously and flexibly on changing operation conditions
- Existing standards and approaches
  - Like VDI-guideline 2206 ‘‘Design methodology for mechatronic systems” or engineering design by Pahl/Beitz (Pahl et al.2007)
  - are no longer adequate
  - especially in the early design phases of ‘‘planning and clarifying the task’’ and ‘‘conceptual design’’ which result in the so-called ‘‘principle solution’’
    - determines the basic structure and the operation mode of the systems
    - subsequently, it is the basis for further concretization
- We will discuss a specification technique, which is able to specify the principle solution for (self-optimizing) mechatronic systems

7 RailCab – Autonomous Railway Transportation System

(University of Paderborn)

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- Replace conventional trains with small, intelligent, demand-driven rail vehicles — autonomous RailCabs that form dynamic convoys on shared track infrastructure.
- Autonomous operation
- Distributed control



IV.1.1 Motivation

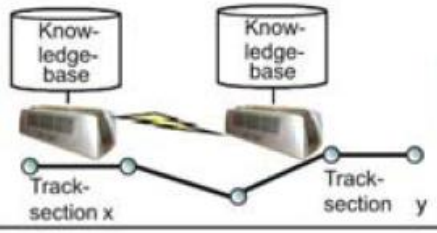
Overview of the partial models described by CONSES

Requirements

2	Geometry
2.1	Length l_{ges}: 6600 mm
2.2	Width w_{ges}: 2420 mm
2.3	Height h_{ges}: 2855 mm
2.4	Distance d_{Rs}: >400 mm
2.5	Distance b_{Spur}: 1435 mm

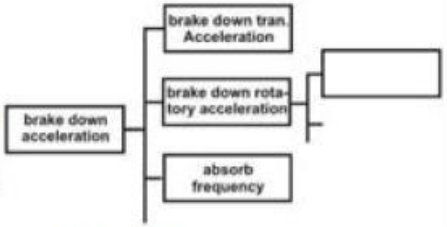
1.

Application Scenarios



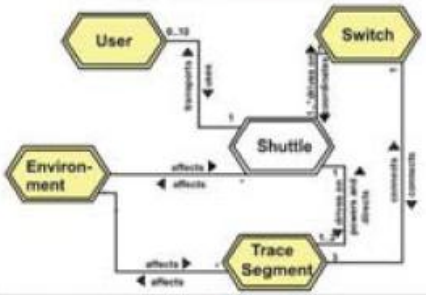
2. Real situations

Functions



3. Must do (hierarchy)

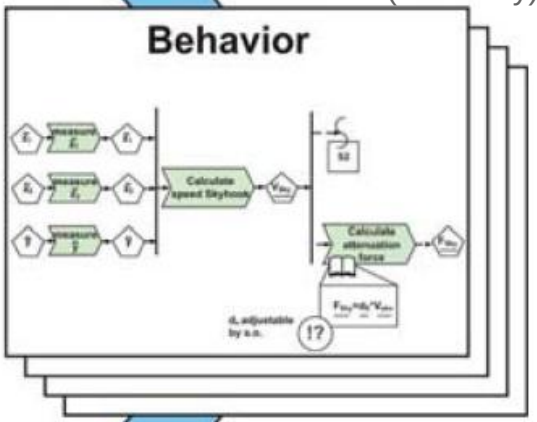
Environment



8.

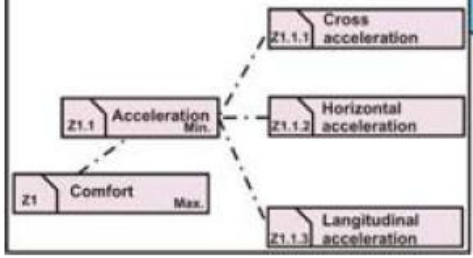
System of Coherent Partial Models

Behavior

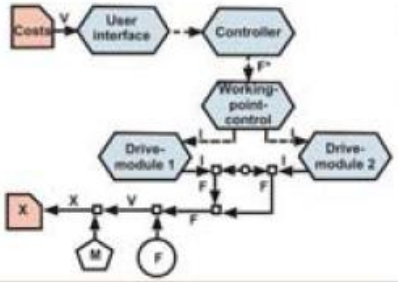


4. Behaves dynamically over time

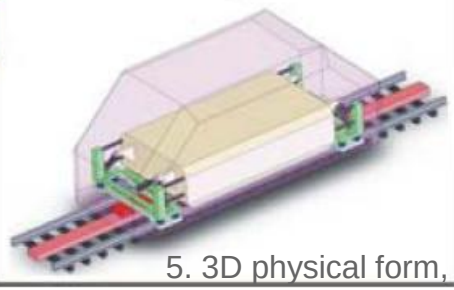
System of Objectives



Active Structure



Shape



5. 3D physical form, geometry

7. Opt targets, goals,

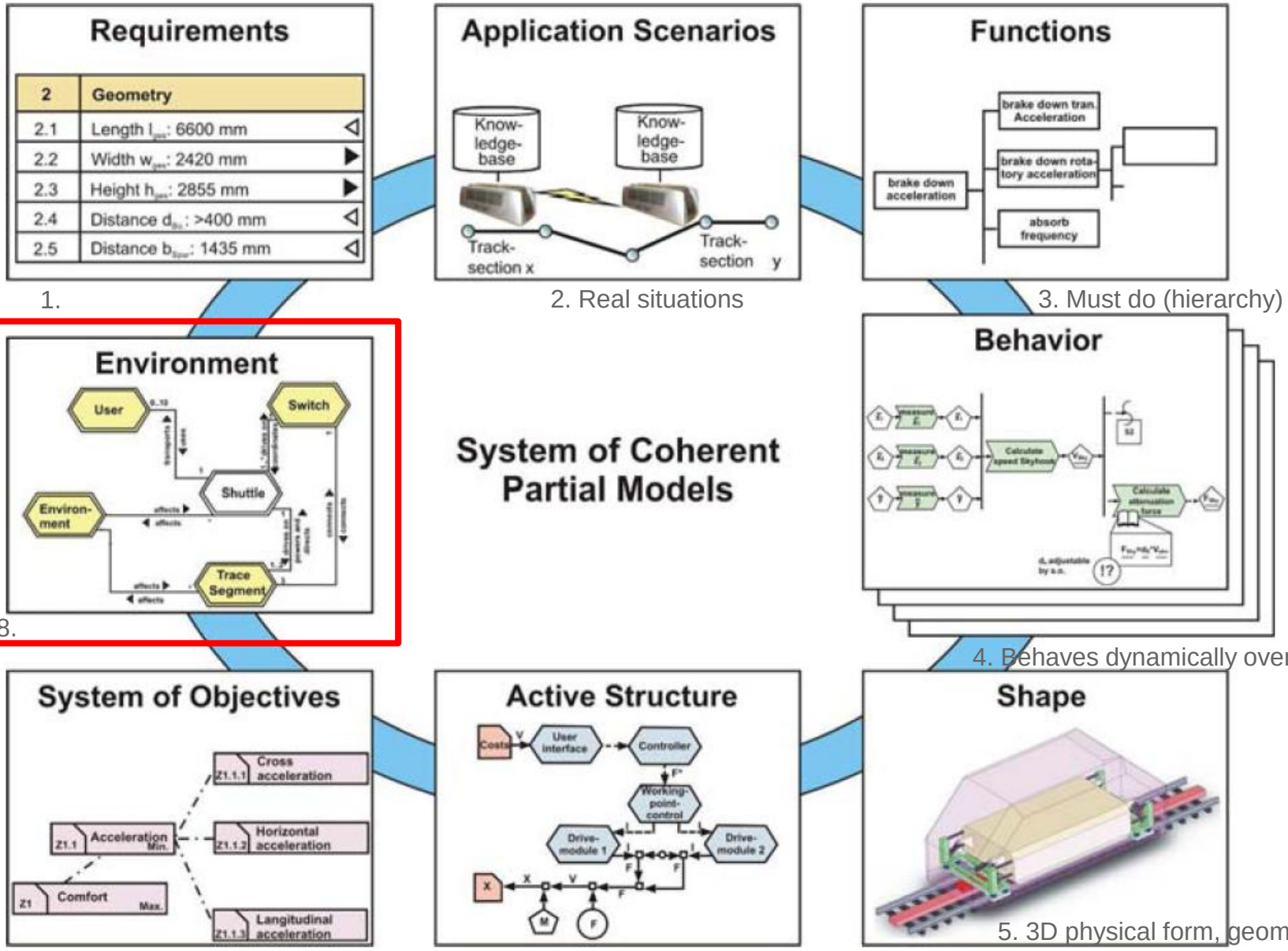
6. physical/logical components, connections

Motivation

- The **environment** of mechanical systems is of vital importance
  - Embedded Systems interact with the environment through sensors and actuations
- Parts of the **relevant environment** could be
  - Interactions with external systems and human users
  - Properties of external systems and human users,
  - Additionally laws, regulations, standards
- The environment **constrain** the system to be developed
  - It is necessary to document the information of the environment
- Explicit environment information
  - Can aid in guiding the developers in eliciting necessary and relevant information
  - Necessary to identify conflicts
    - between assumptions and facts in the systems's specification
    - Between stakeholder intentions

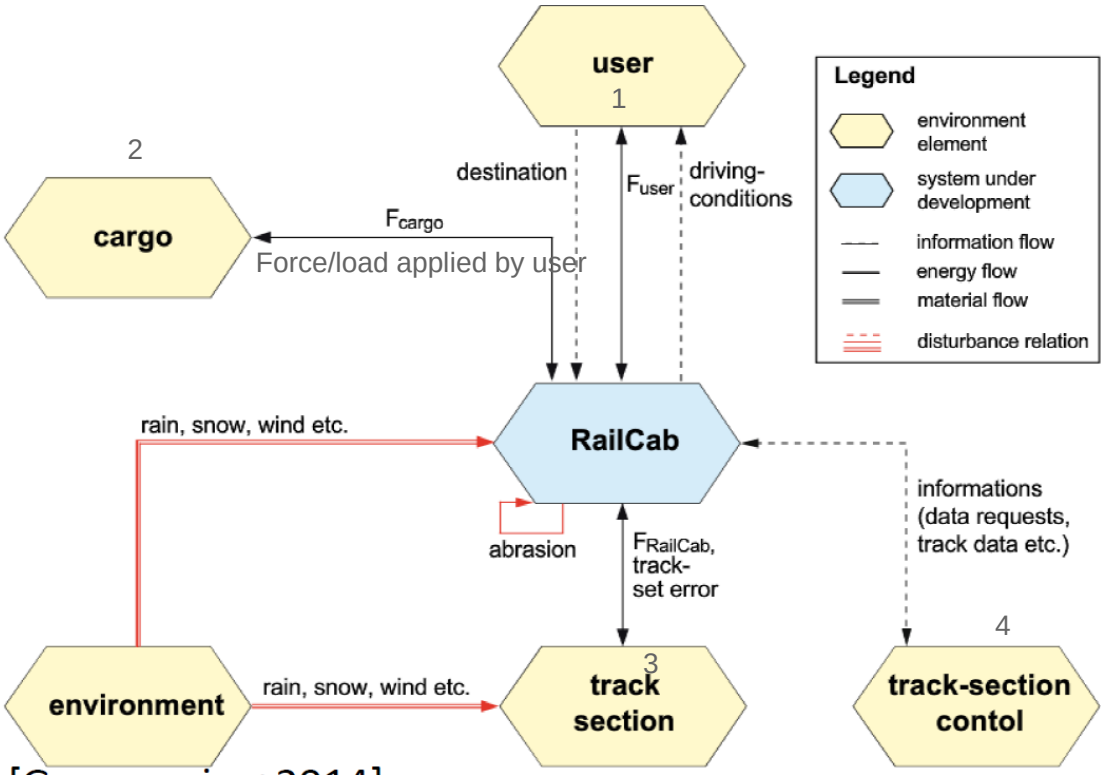
IV.1.1 Motivation

Overview of the partial models described by CONSES



- Environment model of the RailCab

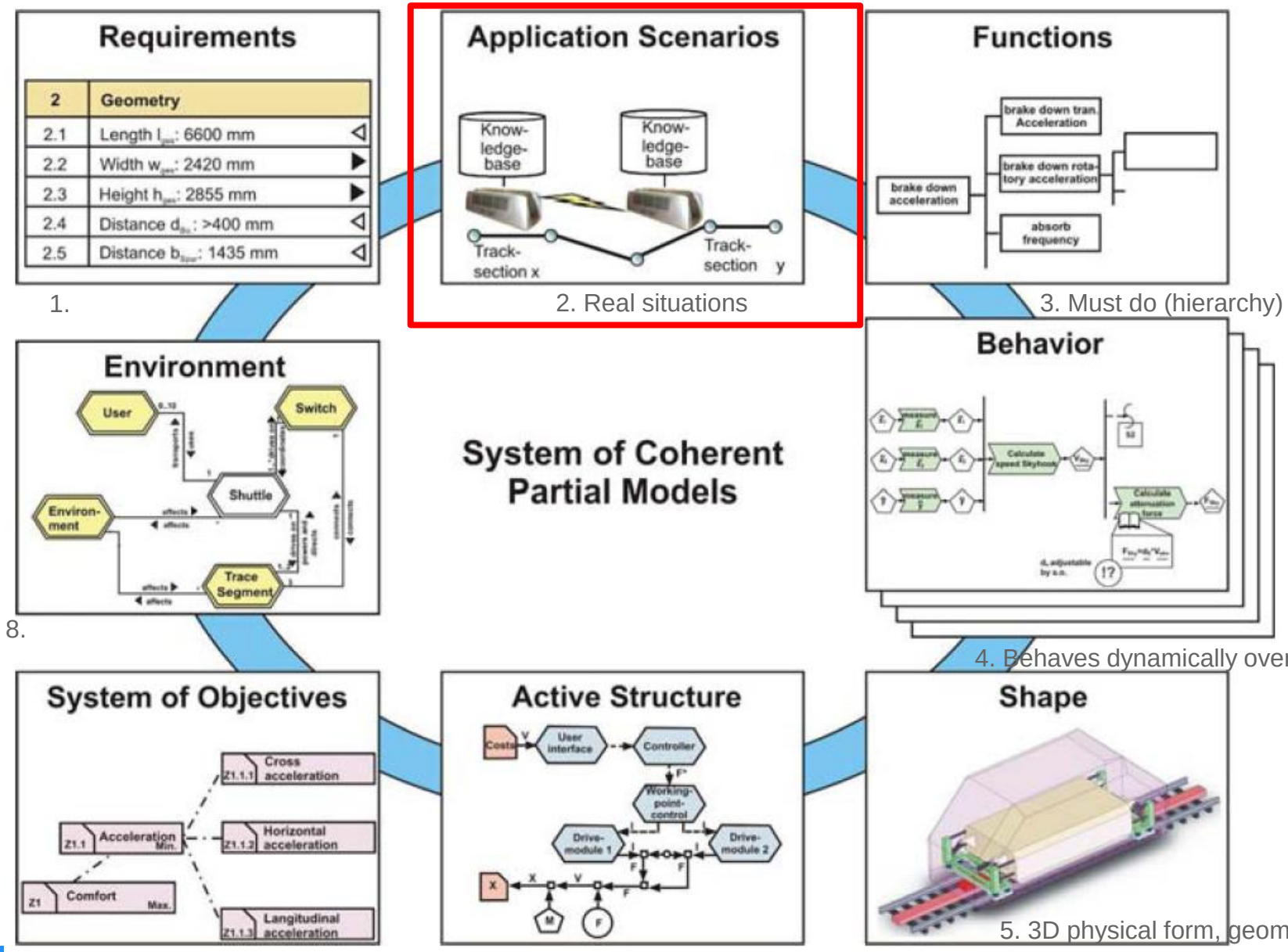




[Gausemeier+2014]

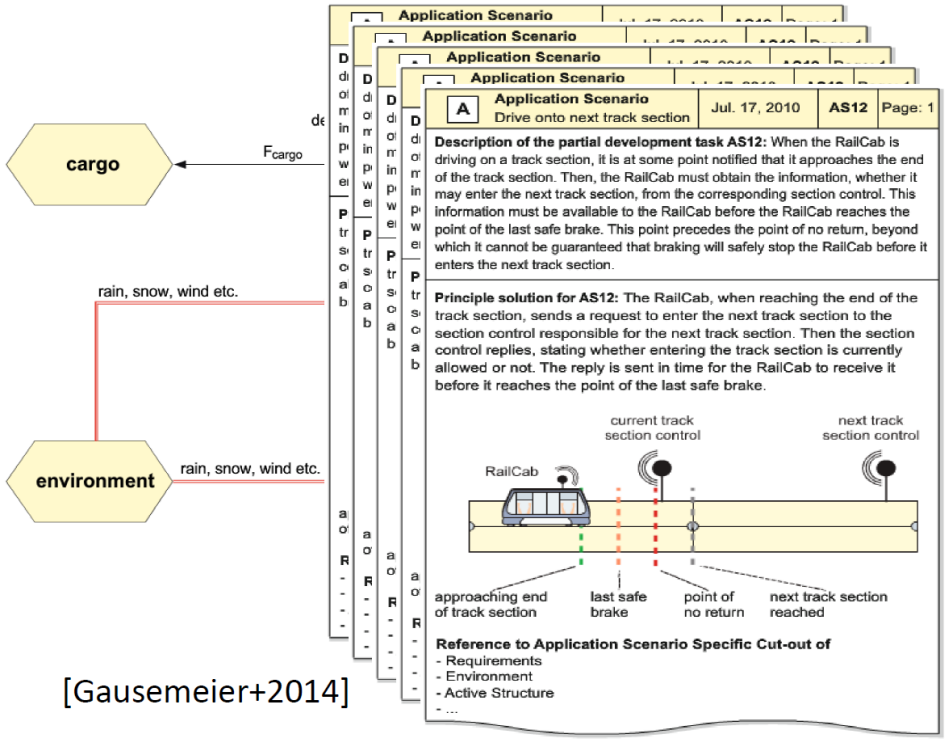
IV.1.1 Motivation

Overview of the partial models described by CONSES



13 Environment and usage analysis

By applying application scenarios



[Gausemeier+2014]

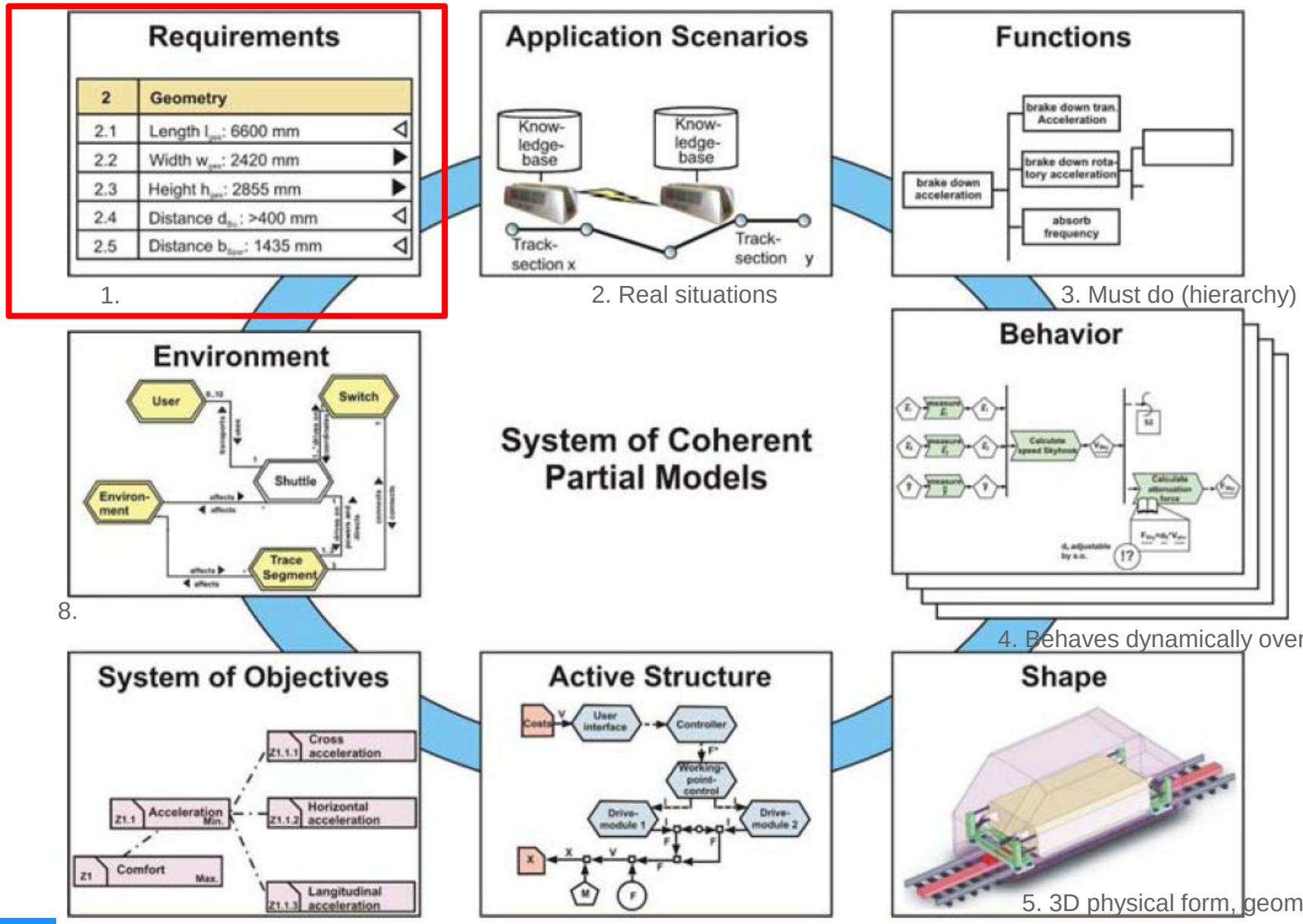
Results:

- Consistent interface definition
- Common definition of the system boundary for all involved disciplines
- Clear development task
- Aligned, understandable description of the Functionality from the customer‘s perspective.

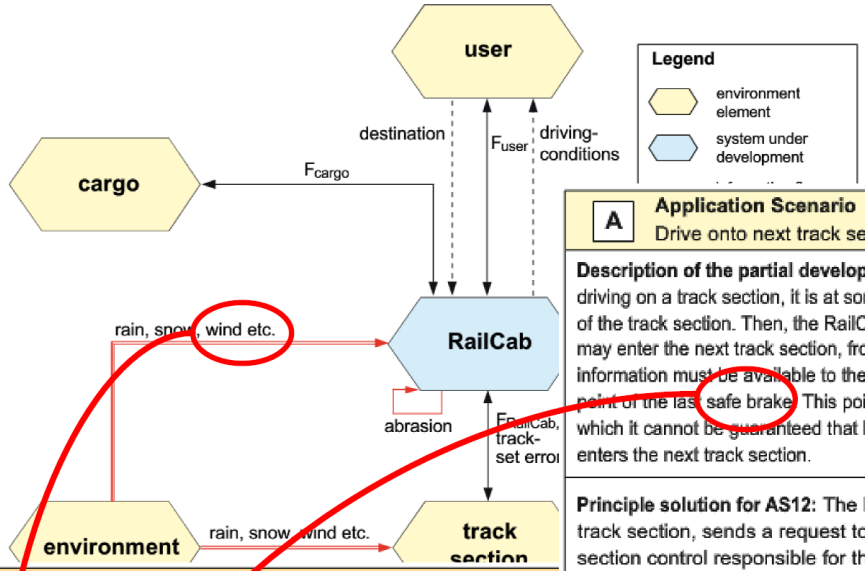
- 1) partial development task: it is approaching the end od the track section, must communicated with next section’s controller before it reaches the point of no return
- 2)Solution: it sends a request to controller, controller (YES/NO), then it acts accordingly.

IV.1.1 Motivation

Overview of the partial models described by CONSES



Requirements definition

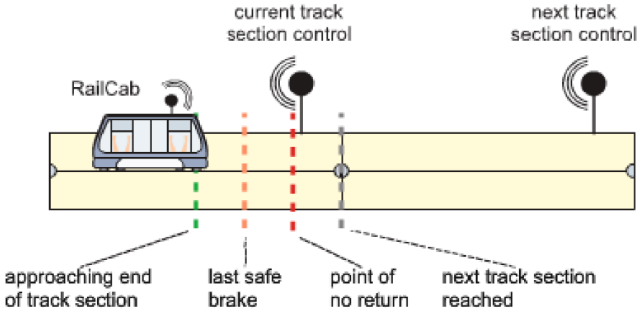


Requirements list	
No.	**Requirement description**
1	Geometry
...	...
1.9	Entrance should be possible from both sides.
1.10	Optimal aerodynamics for single and convoi drive modes.
1.11	Modular construction.
...	...
2	Kinematics
2.1	The vehicle has a steering system.
...	...
7	Safety
...	...
7.9	Provide emergency mechanisms and exits.
7.10	Minimize sensitivity to the side wind.

A	Application Scenario Drive onto next track section	Jul. 17, 2010	AS12	Page: 1

Description of the partial development task AS12: When the RailCab is driving on a track section, it is at some point notified that it approaches the end of the track section. Then, the RailCab must obtain the information, whether it may enter the next track section, from the corresponding section control. This information must be available to the RailCab before the RailCab reaches the point of the last safe brake. This point precedes the point of no return, beyond which it cannot be guaranteed that braking will safely stop the RailCab before it enters the next track section.

Principle solution for AS12: The RailCab, when reaching the end of the track section, sends a request to enter the next track section to the section control responsible for the next track section. Then the section control replies, stating whether entering the track section is currently allowed or not. The reply is sent in time for the RailCab to receive it before it reaches the point of the last safe brake.



Reference to Application Scenario Specific Cut-out of
- Requirements
- Environment
- Active Structure
- ...

Results:

- Clear and traceable origin of the requirements (Traceability)
- Categorization of requirements
- Definition of functional and non-functional requirements in a single document
- Derivation of system and sub-system functions from functional requirements.

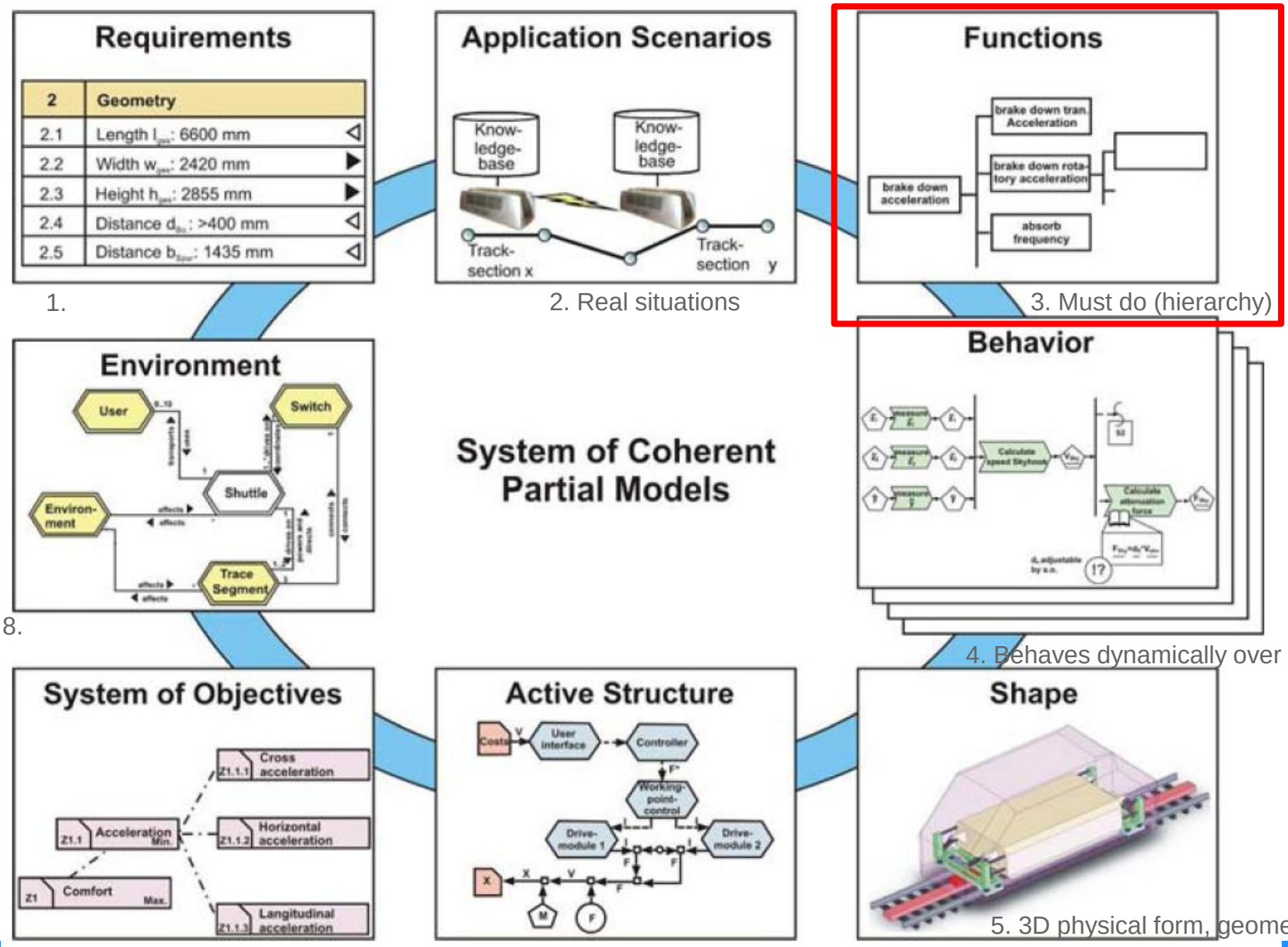
Traceability: why does this req exist?

Where did it come from?

Safety critical systems

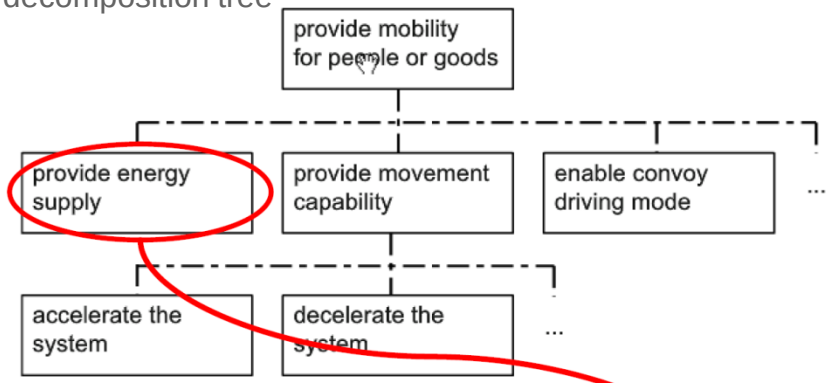
IV.1.1 Motivation

Overview of the partial models described by CONSES

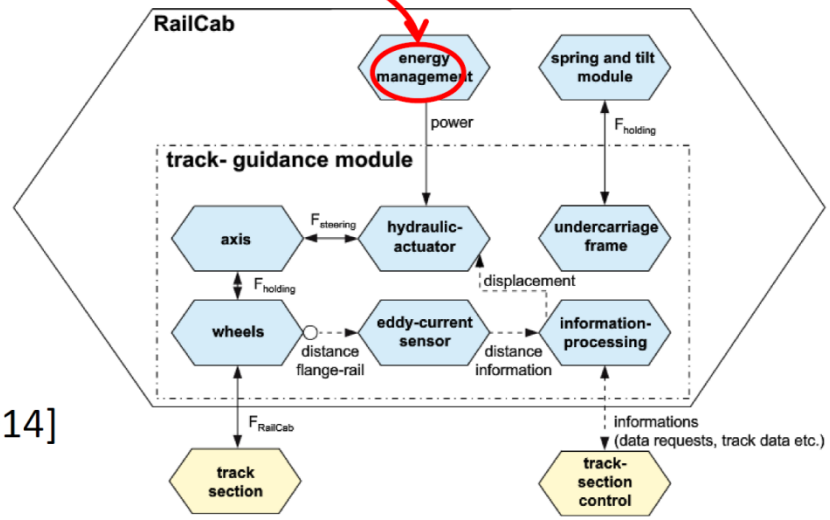


Development of function hierarchy and the active structure

Functional decomposition tree



Active structure



[Gausemeier+2014]

Results:

- Hierarchical order of system functions
- Assignment of solution elements to the functions
- System architecture with solution elements which are connected through material-, energy-, and information-flow
- Foundation for hierarchization and modularization of the system.

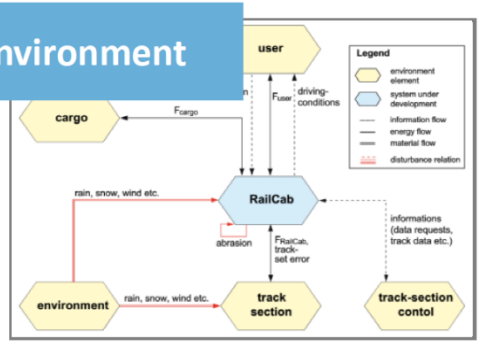
Procedure to develop a product conception

System model with different views

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Who/what surrounds the system

Environment



2. Constraint

What must do, what are the constraints?

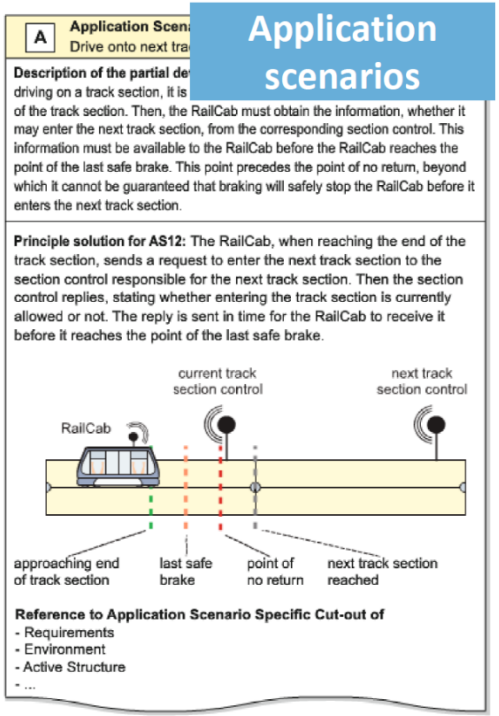
Requirements

No.	Requirement descr	
1	Geometry	
...	...	
1.9	Entrance should be possible from both sides.	D
1.10	Optimal aerodynamics for single and convoi drive modes.	D
1.11	Modular construction.	D
...	...	...
2	Kinematics	D
2.1	The vehicle has a steering system.	D
...	...	...

1. identify

Specific situations must be handeled

Application scenarios

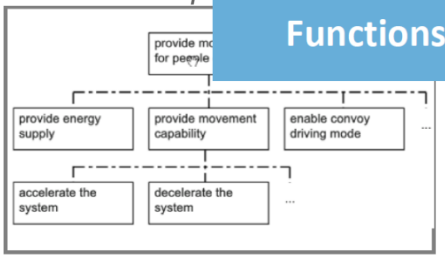


[Gausemeier+2014]

3. Derived from

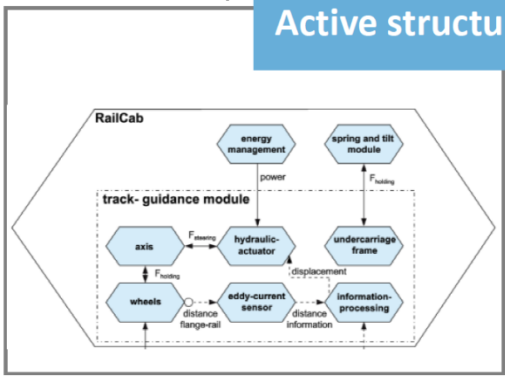
Tasks decomposition

Functions



4. implements

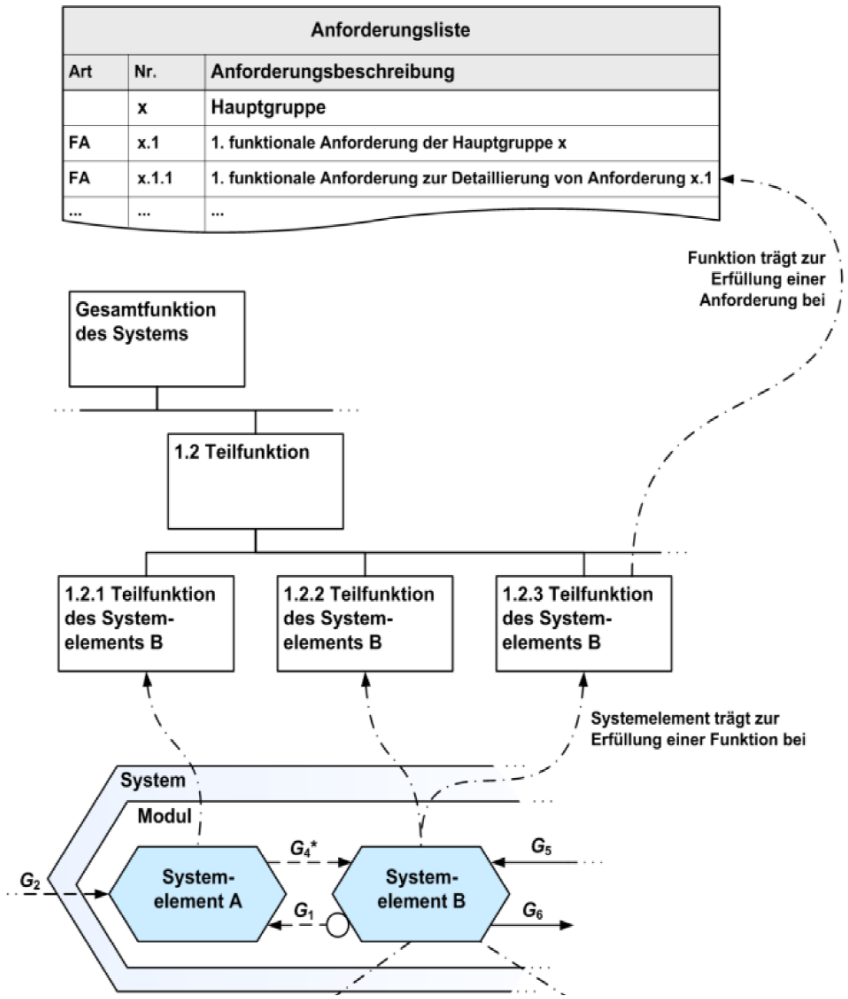
Active structure



Which components implement the functions

Cross-model relation

Traceability



Requirements

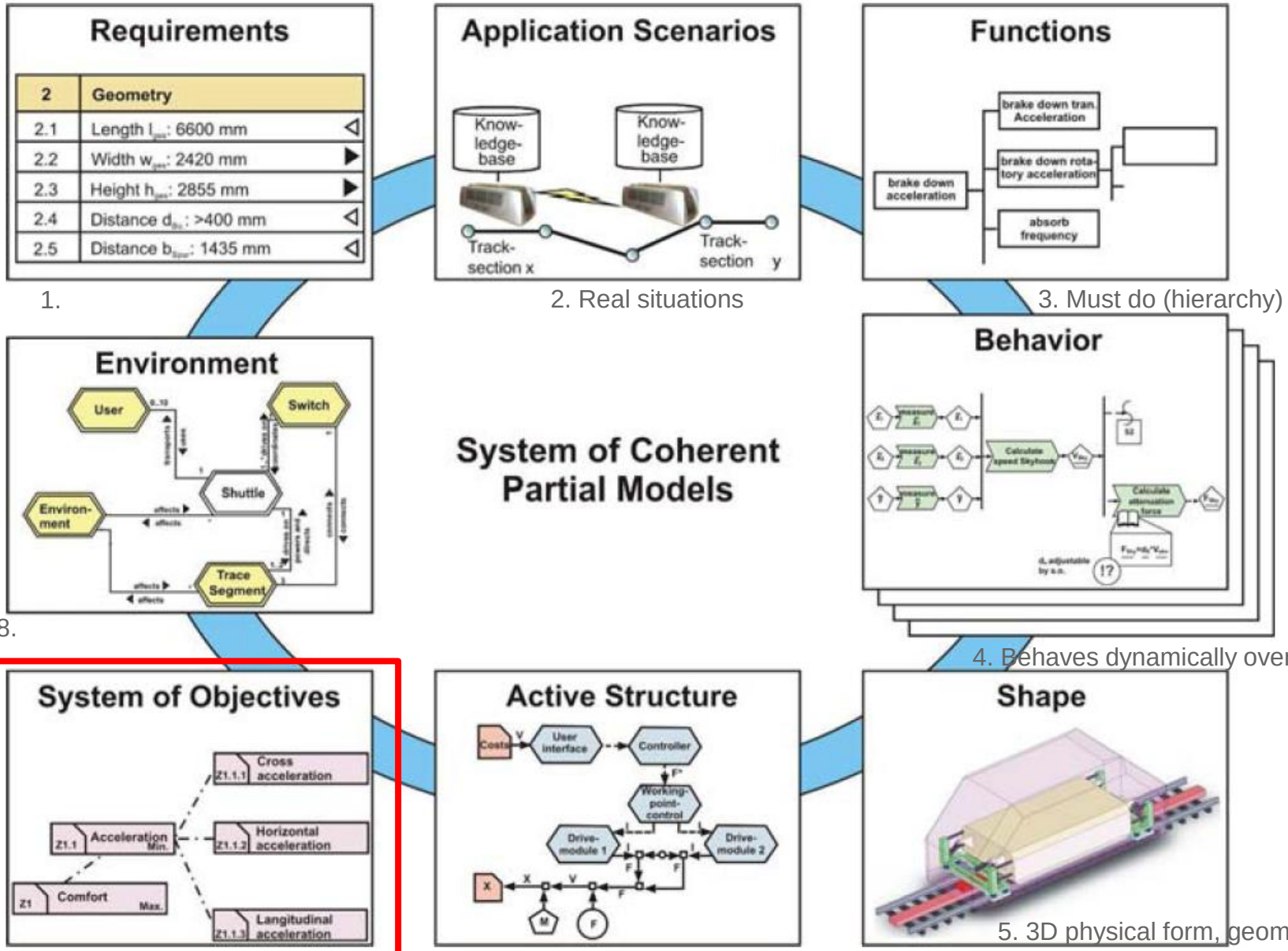
Functions

Active Structure

Safety critical systems: regulators want to see every requirement implemented somewhere in the system, and every component exists for a reason.

IV.1.1 Motivation

Overview of the partial models described by CONSES



21 System of Objectives

Formalize what the system is trying to optimize

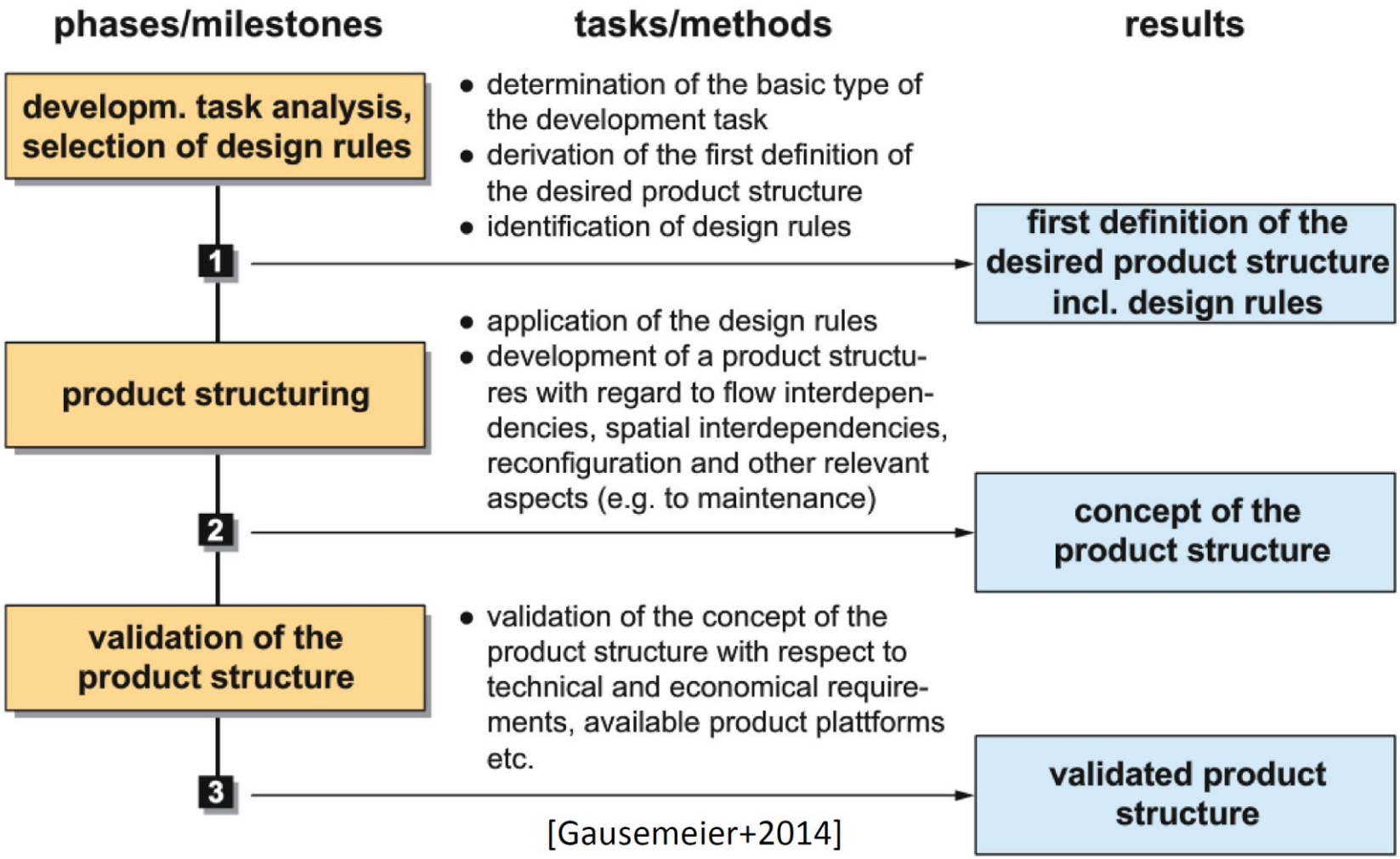
Internal System of Objectives Shuttle (Cut-out)

- O9.1 Lateral Acceleration min
- O9.2 Costs min
- O9.3 Travelling Time min
- O9.4 Abrasion min
- O9.5 Maintenance Interval max

Correlation of the Objectives (Cut-out)

	O9.1	O9.2	O9.3	O9.4	O9.5	...
O9.1						
O9.2	-					
O9.3	-	-				
O9.4	o	o	-			
O9.5	o	o	-	+		
...						

22 Process of product structuring



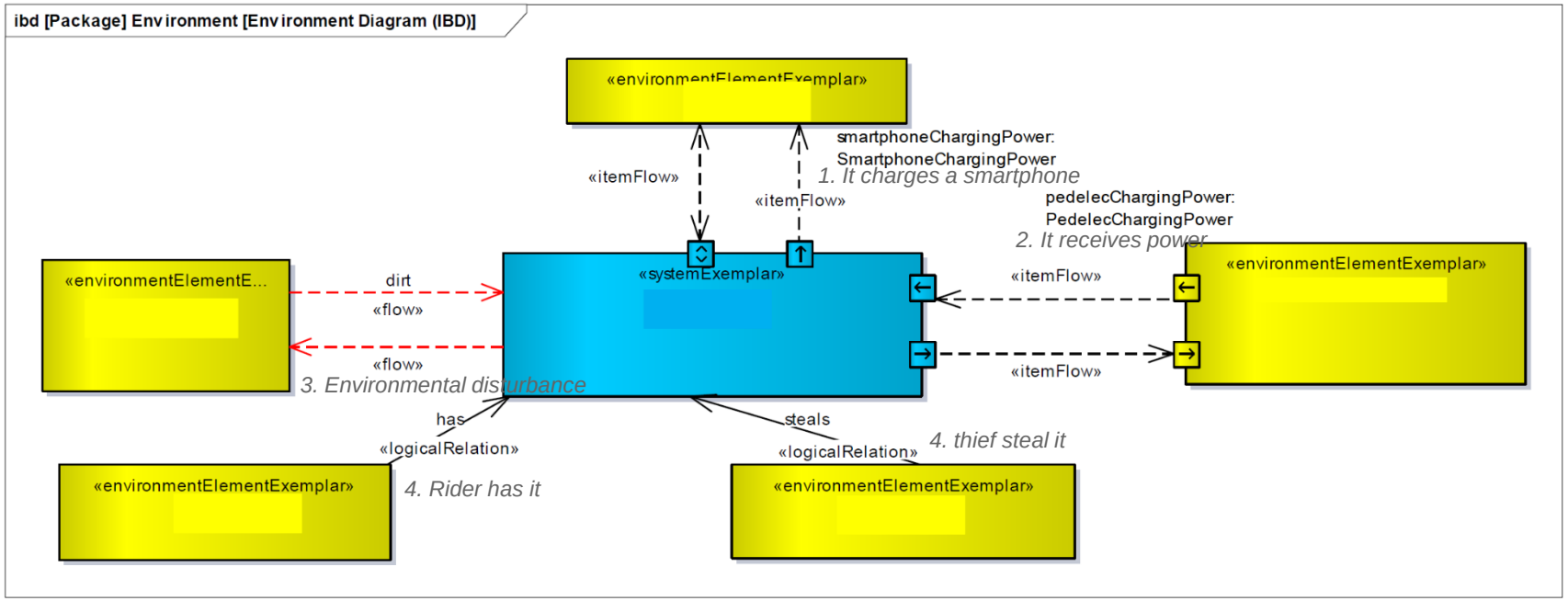
[Gausemeier+2014]

Modeling by using EA / SysML

- If possible, the visual syntax of SysML can be used to specify the different views of the CONSENS approach
- Enterprise Architect (EA) is a possible tool

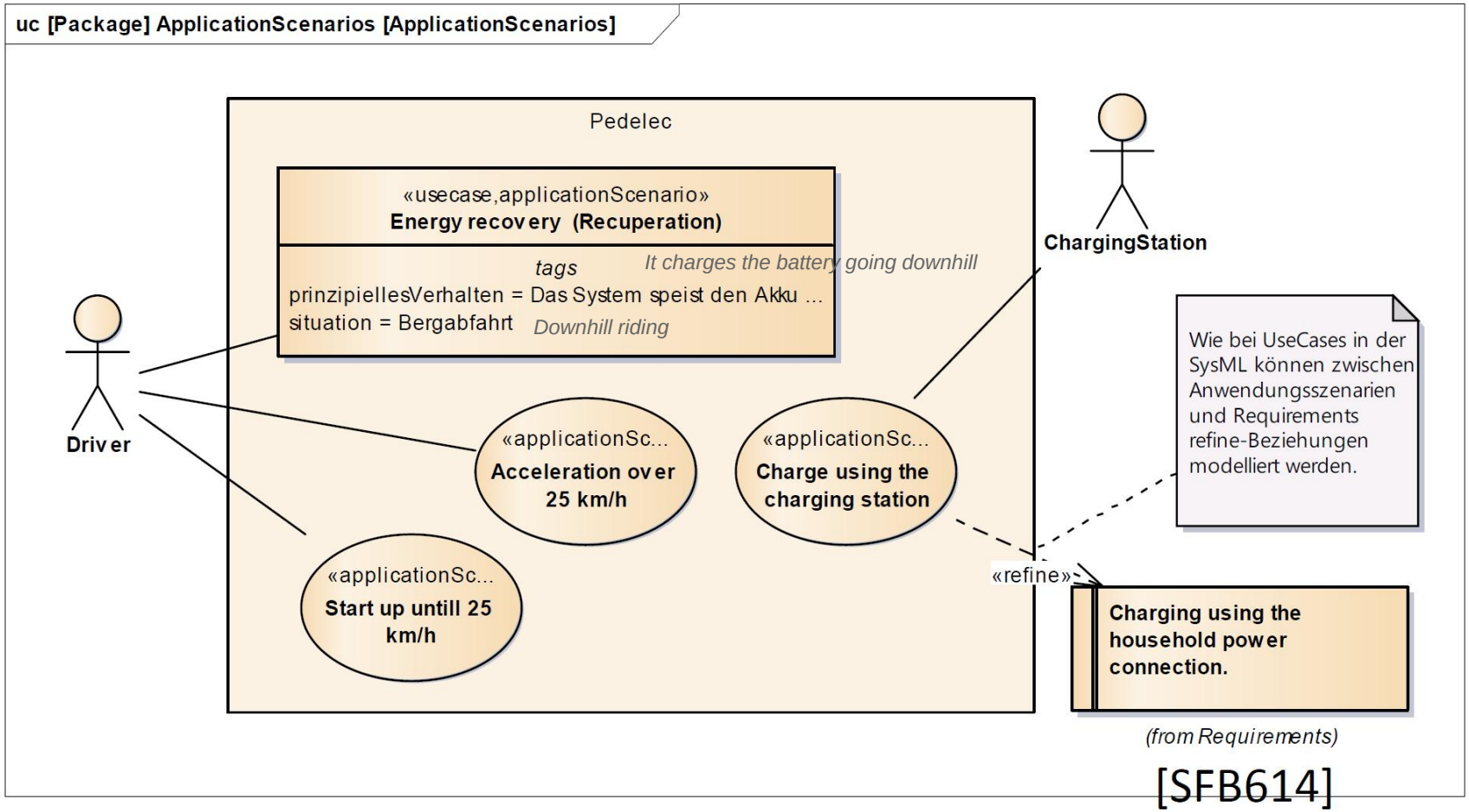
- In the following:
  - Pedelec example (an electric assisted biecycle)



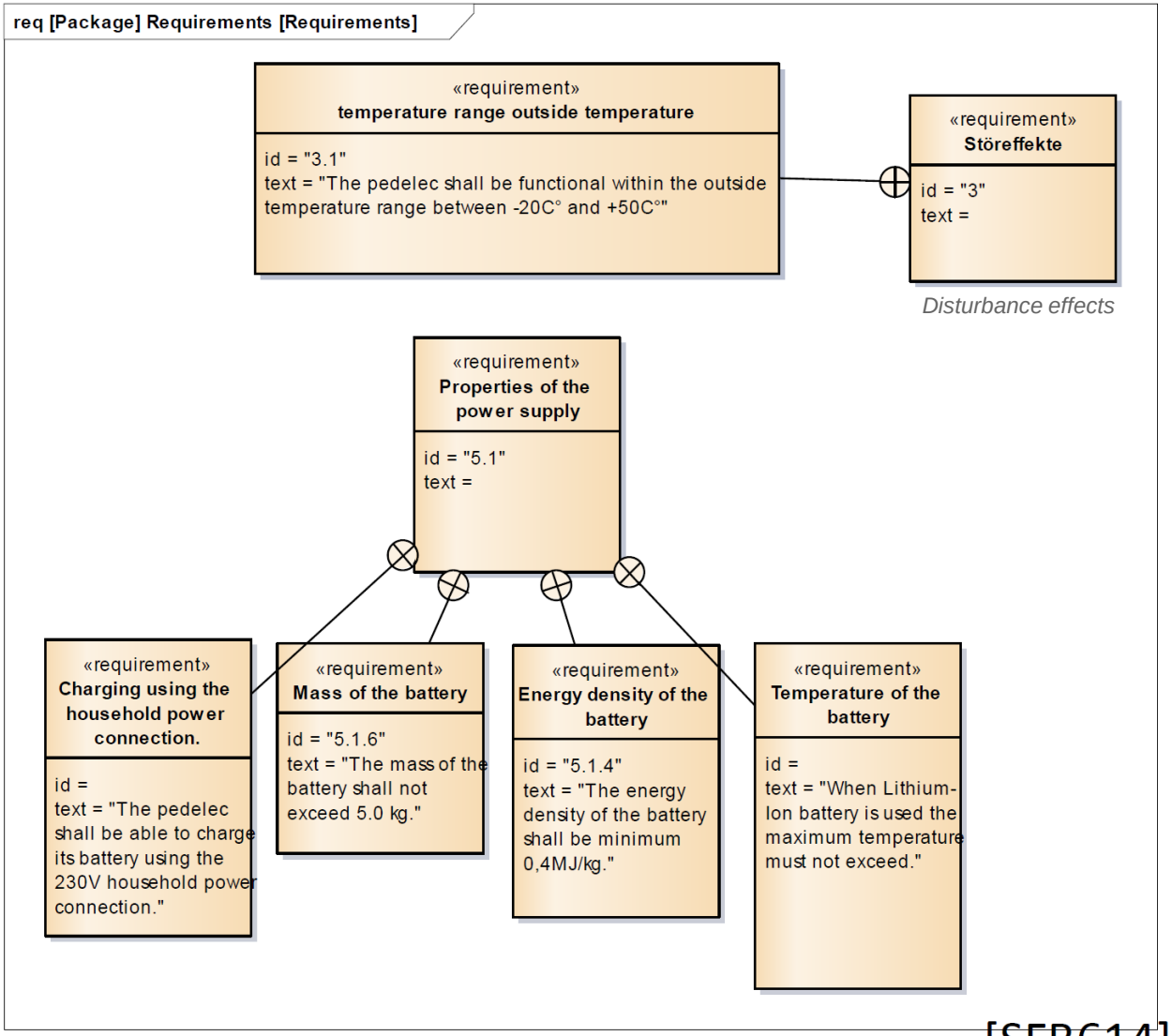

ibd [Package] Environment [Environment Diagram (IBD)]
«environmentElementExemplar»
smartphoneChargingPower:
SmartphoneChargingPower
«itemFlow»
«itemFlow»
1. It charges a smartphone
pedelecChargingPower:
PedelecChargingPower
2. It receives power
«environmentElementE...
dirt
«flow»
«flow»
«systemExemplar»
«itemFlow»
«itemFlow»
«environmentElementExemplar»
3. Environmental disturbance
has
«logicalRelation»
steals
«logicalRelation»
4. thief steal it
«environmentElementExemplar»
4. Rider has it
«environmentElementExemplar»

[SFB614]

Application Scenarios



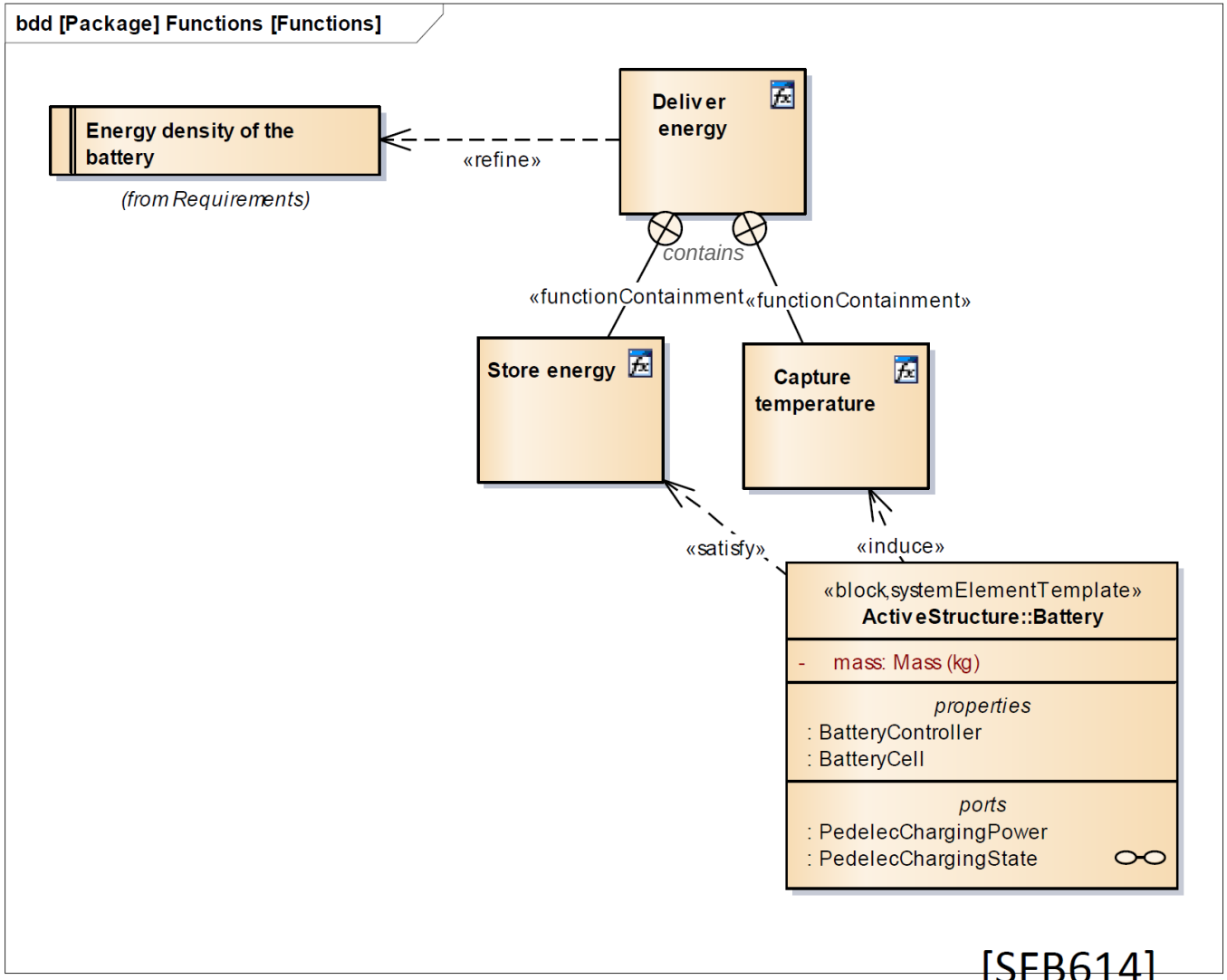
Modeling requirements



[SFB614]

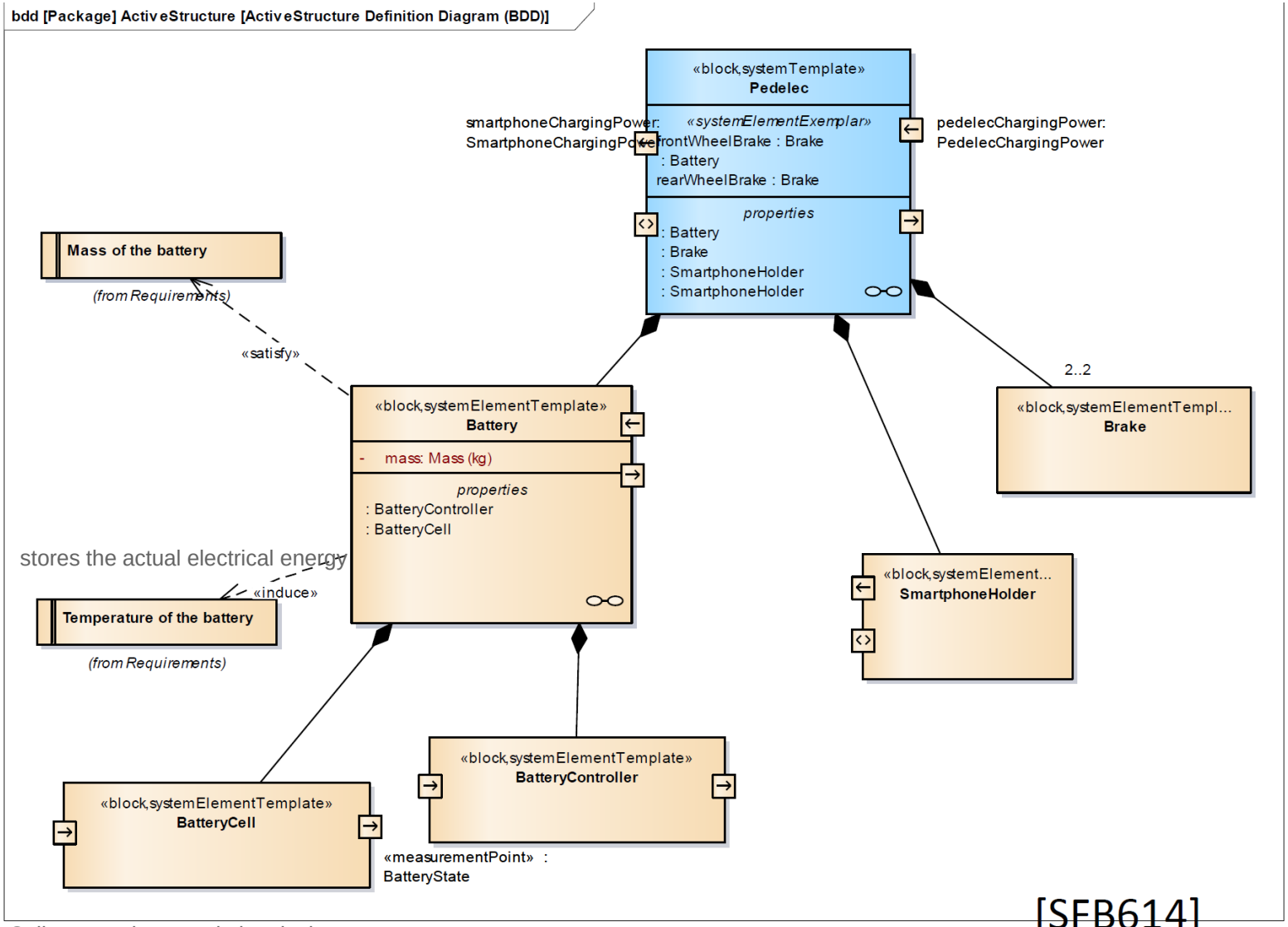
Modelling Functions

Pedal battery subsystem



[SFB614]

Modeling Active Structure



[SFB614]

BatteryCell: stores the actual electrical energy

BatteryController — manages and monitors the battery, with a *BatteryState* measurement point so the system always knows the current status

Summary

- The specification technique **CONSENS is strongly interconnected** with the reference process and focuses on self-optimizing mechatronic systems **than other approaches like UML and SysML.**

30 References

- [Gausemeier+2014, Cha 4.1]Design Methodology for Intelligent Technical Systems ,Develop Intelligent Technical Systems of the Future.
- [J.Gausemeier , U.Frank , J.Donoth,S.Kahl, 2009] Specification technique for the description of self-optimizing mechatronic systems